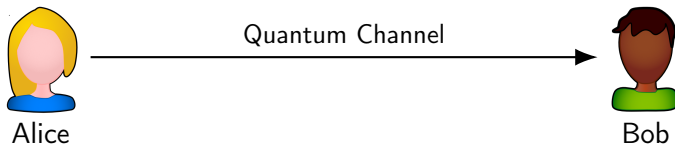


Quantum Cryptography : Part 3

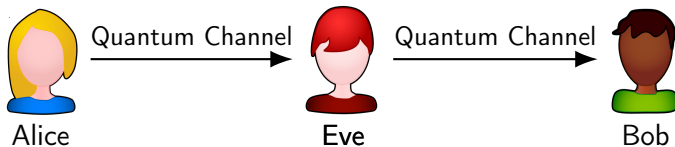
Eve, Detection & Future

“The attacker becomes part of the experiment.”

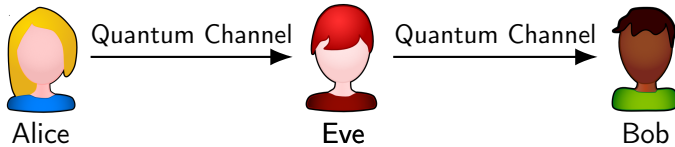
Eve Enters the Channel



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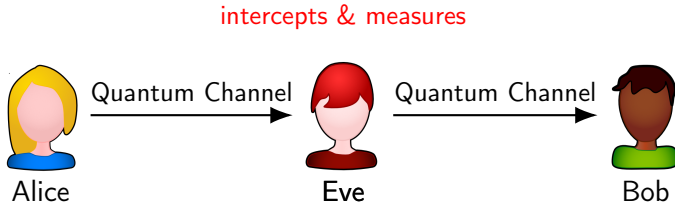


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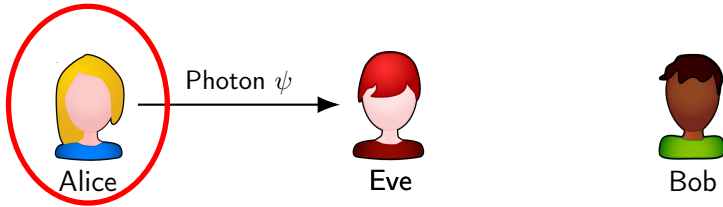
Eve sits in the path, wanting the key — with one option: measure it

Eve Enters the Channel



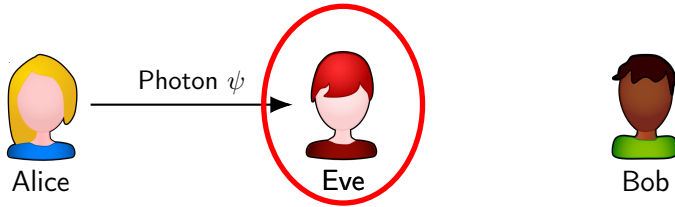
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Intercept-Measure-Resend



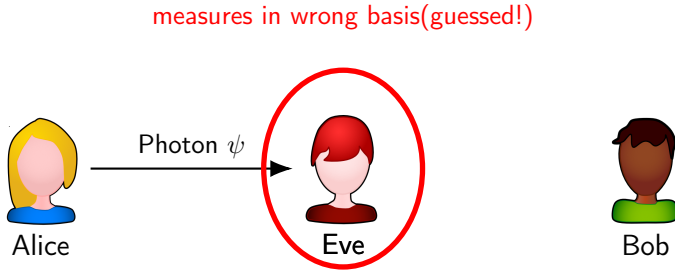
A photon leaves Alice, encoded in her random basis.

Intercept-Measure-Resend



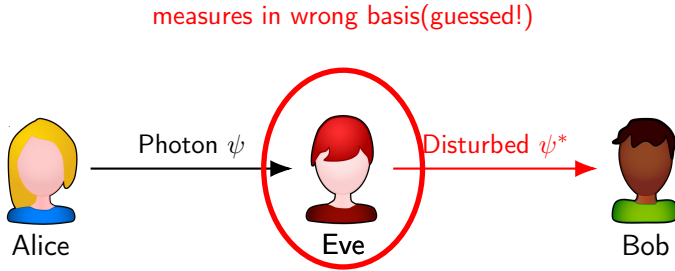
Eve intercepts it mid-flight

Intercept-Measure-Resend



She must guess the basis — half the time, she's wrong.

Intercept-Measure-Resend



Wrong guess = altered state sent to Bob. The evidence just changed.

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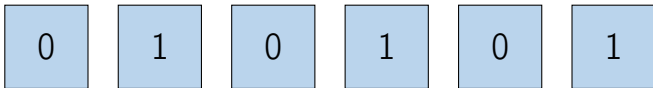
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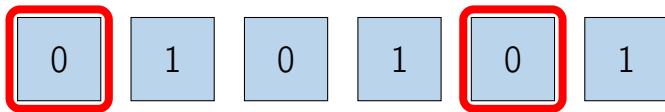
That's not a bug — that's her fingerprint.

Catching Eve: Sampling the Key



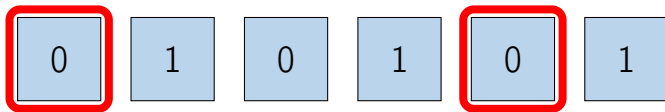
A small random sample of the sifted key is picked.

Catching Eve: Sampling the Key



sacrificed publicly to check for tampering

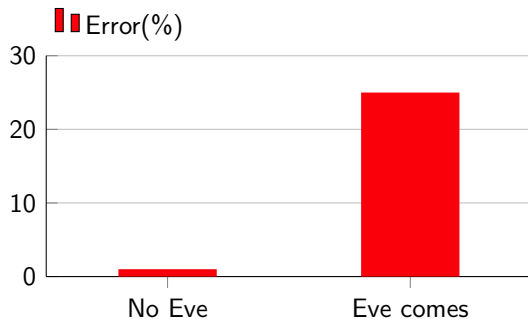
Catching Eve: Sampling the Key



sacrificed publicly to check for tampering

If nobody touched the channel, these should match exactly.

The Error Rate Tells the Story



Near zero without an eavesdropper — a clear jump to $\sim 25\%$ with one. Unmissable.

Bit-by-Bit: Where They Disagree

Alice	0	1	0	1	0	1
Bob	0	1	1	1	1	1

Compare Alice's and Bob's sampled bits directly.

Bit-by-Bit: Where They Disagree

Alice	0	1	0	1	0	1
Bob	0	1	1	1	1	1

Every mismatch is Eve's signature — written in physics.

Classical vs. Quantum Eavesdropping

Classical

- ▶ Tap a wire, copy the signal
- ▶ The original is untouched
- ▶ No trace left behind

Quantum

- ▶ Measuring changes state
- ▶ Original is disturbed
- ▶ Detected, every time

Eavesdropping stops being invisible — it becomes measurable.

Detection

Comparing the sampled bits. . .

Detection

EVE DETECTED!!!

Detection

EVE DETECTED!!!

Not suspected. Not guessed.
Detected — by the laws of physics.

Beyond the Lab

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“What if looking at a secret could change it?”

Now you know: it does. And that's what makes it unbreakable.

Quantum Cryptography : Part 3

Thank You!